BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Topology of Metric Spaces

Open sets, Limit points and Closed sets

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Conducted by

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Open and Closed Spheres

Open Sphere

Let (X, d) be a metric space and let $a \in X$. For any $r > 0$, the set

$$S_r(a) = \{x \in X : d(x, a) < r\}$$

is called an *open sphere* (or *open ball*) of radius r centered at a.

Closed Sphere

Let (X, d) be a metric space and let $a \in X$. For any $r > 0$, the set

$$S_r[a] = \{x \in X : d(x, a) \leq r\}$$

is called a *closed sphere* of radius r centered at a.

Relation Between Open and Closed Spheres

Show that

$$S_r(a) \subseteq S_r[a]$$

for every $a \in X$ and for every $r > 0$.

Neighbourhood of a Point

Neighbourhood of a Point

Let (X, d) be a metric space and let $a \in X$. A subset $N_a \subseteq X$ is called a *neighbourhood* of the point a if there exists an open sphere $S_r(a)$ centered at a such that

$$S_r(a) \subseteq N_a$$

for some $r > 0$.

Open Sphere Is a Neighbourhood of Its Points

Show that every open sphere is a neighbourhood of each of its points.

Open Set

Open Set

A subset G of a metric space (X, d) is said to be *open* in X (with respect to the metric d) if G is a neighbourhood of each of its points. In other words, G is open if for each $a \in G$ there exists an $r > 0$ such that

$$S_r(a) \subseteq G.$$

Illustrations of Open Sets

Show that each of the following subsets is open.

1. The empty set $\emptyset$ and the entire space X with any metric are open sets.
2. Every open sphere in a metric space (X, d) is an open set.
3. Let $A = \{z \in \mathbb{C} : 1 < |z| < 2\}$ with the usual metric. Then A is an open set.
4. Let
$$S = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$$
with the Euclidean metric. Then S is an open set.
5. Let
$$A = \{(x, y) \in \mathbb{R}^2 : y^2 < x\}$$
with the Euclidean metric. Then A is an open set.
6. Let
$$A = \left\{f \in C[a, b] : \inf_{x \in [a,b]} f(x) > 0\right\}.$$
Then A is open in $C[a, b]$ with respect to the supremum metric.

Singleton Set in $\mathbb{R}$ Is Not Open

Show that on the real line with the usual metric, the singleton set $\{x\}$ is not open.

Singleton Sets Under Different Metrics

Let $\mathbb{R}$ be the set of real numbers, d the usual metric and d' the discrete metric on the same set $\mathbb{R}$. Show that every singleton set $\{x\}$, $x \in \mathbb{R}$, is open in $(\mathbb{R}, d')$ but not open in $(\mathbb{R}, d)$.

Discrete Metric Space: Every Set Is Open

Show that every subset of a discrete metric space (X, d) is open.

A Subset Open in a Subspace

Let (X, d) be a metric space where $X = (0, 2)$ with the usual metric. Show that the subset

$$A = (0, 1)$$

is an open set in X.

Basic Properties of Open Sets

In any metric space (X, d),

1. the union of an arbitrary family of open sets is open;

2. the intersection of a finite number of open sets is open.

Intersection of Infinite Open Sets May Not Be Open

Show that the intersection of an infinite number of open sets in a metric space need not be open.

Limit Point

Limit Point

Let (X, d) be a metric space and let $A \subseteq X$. A point $x \in X$ is called a *limit point* (or *accumulation point*) of the set A if for every $r > 0$,

$$S_r(x) \cap (A \setminus \{x\}) \neq \emptyset.$$

Illustrations of Limit Points and Derived Sets

Determine the set of limit points (derived set) in each of the following cases.

1. Let $X = \mathbb{R}$ with the usual metric and let

$$A = [0, 1).$$

   Every point of A is a limit point of A, and 1 is also a limit point of A although $1 \notin A$. Hence

$$A' = [0, 1].$$

2. Let

$$A = \left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$$

   with the usual metric in $\mathbb{R}$. Then 0 is the only limit point of A and $0 \notin A$. Hence

$$A' = \{0\}.$$

3. If (X, d) is a discrete metric space, then the derived set of every subset of X is empty, that is,

$$A' = \emptyset \quad \text{for all } A \subseteq X.$$

4. Every real number is a limit point of the set of rational numbers $\mathbb{Q}$. Hence

$$\mathbb{Q}' = \mathbb{R}.$$

5. The set of integers $\mathbb{Z}$ has no limit point, that is,

$$\mathbb{Z}' = \emptyset.$$

6. Every finite subset of a metric space has no limit point.
7. Let
$$A = \{z \in \mathbb{C} : 1 < |z| < 2\}$$
with the usual metric. Then
$$A' = \{z \in \mathbb{C} : 1 \leq |z| \leq 2\}.$$

Closed Sets

Closed Set

A subset F of a metric space (X, d) is said to be *closed* if F contains all its limit points.

Illustrations of Closed Sets

Verify the following statements concerning closed sets in metric spaces.

1. The empty set $\emptyset$ and the whole space X are closed sets in every metric space (X, d).
2. On the real line $\mathbb{R}$ with the usual metric, the set $\mathbb{N}$ of natural numbers is closed.
3. The set $\mathbb{Q}$ of rational numbers is not closed in $\mathbb{R}$ with the usual metric.
4. Every closed interval on the real line is a closed set.
5. Every finite subset of a metric space is closed.
6. Let
$$A = \{f \in C[a, b] : f(a) = 1\}.$$
Then A is closed in $C[a, b]$ with respect to the supremum metric.
7. The set
$$\left\{\frac{1+i}{n} : n \in \mathbb{N}\right\}$$
is neither open nor closed in the complex plane with respect to the usual metric.

Equivalent Definition of Closed Sets

Let (X, d) be a metric space. A subset $F \subseteq X$ is closed if and only if its complement $X \setminus F$ is open.

Closed Sphere Is a Closed Set

Show that every closed sphere in a metric space (X, d) is a closed set.

Basic Properties of Closed Sets

In any metric space (X, d),

1. the intersection of an arbitrary family of closed sets is closed;

2. the union of a finite number of closed sets is closed.

Union of Infinite Closed Sets May Not Be Closed

Show that the union of an infinite number of closed sets in a metric space need not be closed.

Closure of a Set

Closure

Let A be a subset of a metric space (X, d). The *closure* of A, denoted by $\overline{A}$ is the smallest closed set containing A, that is,

$$\overline{A} = \bigcap \{F : F \text{ is closed and } A \subseteq F\};$$

Equivalent Definition of Closure

Let A be a subset of a metric space (X, d). Then

$$\overline{A} = A \cup A'.$$

Illustrations of Closure

Determine the closure of each of the following subsets.

1. Let $X = \mathbb{R}$ with the usual metric and let
$$A = (0, 1) \cup (1, 2).$$
Then
$$\overline{A} = [0, 2].$$

2. Let $X = \mathbb{R}$ with the usual metric and let
$$A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}.$$
Then
$$\overline{A} = A \cup \{0\}.$$

3. If (X, d) is a discrete metric space and $A \subseteq X$, then
$$\overline{A} = A.$$

4. Let $X = \mathbb{N}$ with the usual metric and let $A = \{1, 2, 3\}$. Then
$$\overline{A} = A.$$

Properties of Closure

Let (X, d) be a metric space and let A, B be any two subsets of X. Then:

1. A is closed if and only if $A = \overline{A}$;
2. $A \subseteq B$ implies $\overline{A} \subseteq \overline{B}$;
3. $\overline{A \cup B} = \overline{A} \cup \overline{B}$;
4. $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.
5. $\overline{\overline{A}} = \overline{A}$.
6. $\overline{A} = (\text{ext}\, A)^c$.
7. $\overline{A} = \text{int}(A^c)^c$.

Interior, Exterior, Frontier and Boundary Points

Interior Point

Let A be a subset of a metric space (X, d). A point $a \in A$ is called an *interior point* of A if there exists $r > 0$ such that

$$a \in S_r(a) \subseteq A.$$

Interior of a Set

The set of all interior points of A is called the *interior* of A and is denoted by $\operatorname{int} A$, where

$$\operatorname{int} A = \{x \in A : S_r(x) \subseteq A \text{ for some } r > 0\}.$$

Clearly, $\operatorname{int} A$ is an open subset of A.

Exterior Point

A point $x \in X$ is called an *exterior point* of A if it is an interior point of the complement of A, that is, if there exists $r > 0$ such that

$$S_r(x) \subseteq A^c \quad \text{or equivalently} \quad S_r(x) \cap A = \emptyset.$$

Exterior of a Set

The set of all exterior points of A, denoted by $\operatorname{ext} A$, is called the *exterior* of A. By definition,

$$\operatorname{ext} A = \operatorname{int} A^c.$$

Hence $\operatorname{ext} A$ is open and is the largest open set contained in A^c. Moreover,

$$\operatorname{ext} A = (\overline{A})^c.$$

Boundary Point

A point $x \in X$ is called a *boundary point* (or *frontier point*) of A if every open sphere centered at x intersects both A and its complement A^c, that is, for every $r > 0$,

$$S_r(x) \cap A \neq \emptyset \quad \text{and} \quad S_r(x) \cap A^c \neq \emptyset.$$

Boundary of a Set

The set of all boundary points of A is called the *boundary* (or *frontier*) of A and is denoted by ∂A, where

$$\partial A = \overline{A} \setminus \operatorname{int} A.$$

Equivalently,

$$\partial A = \overline{A} \cap \overline{A^c}.$$

Illustrations of Interior, Exterior, Boundary

Compute the interior, exterior, and boundary of the following subsets.

1. Let $X = \mathbb{R}$ with the usual metric and let $A = \mathbb{Q}$. Then
$$\operatorname{int} A = \emptyset, \qquad \operatorname{ext} A = \emptyset, \qquad \partial A = \mathbb{R}.$$

2. Let $X = \mathbb{R}$ with the usual metric and let $A = (1,2) \cup (3,4)$. Then
$$\operatorname{int} A = A,$$
$$\operatorname{ext} A = (-\infty, 1) \cup (2,3) \cup (4, \infty),$$
$$\partial A = \{1,2,3,4\}.$$

3. If (X,d) is a discrete metric space and $A \subseteq X$, then
$$\operatorname{int} A = A, \qquad \operatorname{ext} A = A^c, \qquad \partial A = \emptyset.$$

4. Let $X = \mathbb{N}$ with the usual metric and let $A = \{1,2,3\}$. Then
$$\operatorname{int} A = \emptyset, \qquad \operatorname{ext} A = \emptyset, \qquad \partial A = \mathbb{N}.$$

Properties of Interior

Let A and B be any two subsets of a metric space (X, d). Then:

1. $\operatorname{int} A$ is the largest open set contained in A, that is,

$$\operatorname{int} A = \bigcup\{G : G \text{ is open and } G \subseteq A\};$$

2. A is open if and only if $A = \operatorname{int} A$;
3. $A \subseteq B$ implies $\operatorname{int} A \subseteq \operatorname{int} B$;
4. $\operatorname{int}(A \cap B) = \operatorname{int} A \cap \operatorname{int} B$;
5. $\operatorname{int}(A \cup B) \supseteq \operatorname{int} A \cup \operatorname{int} B$.

Properties of Exterior of a Set

Let (X, d) be a metric space and let A, B be any two subsets of X. Then:

1. $\operatorname{ext} A$ is the largest open set contained in A^c;
2. A is closed if and only if $A^c = \operatorname{ext} A$;
3. $A \subseteq B$ implies $\operatorname{ext} B \subseteq \operatorname{ext} A$;
4. $\operatorname{ext}(A \cap B) \supseteq \operatorname{ext} A \cup \operatorname{ext} B$;
5. $\operatorname{ext}(A \cup B) = \operatorname{ext} A \cap \operatorname{ext} B$.

Properties of Boundary

Let (X, d) be a metric space, and let A, B be subsets of X. Then:

1. $\partial A = \overline{A} \cap \overline{A^c} = \overline{A} \setminus \operatorname{int} A$;
2. $\partial A = \emptyset$ if and only if A is both open and closed;
3. A is closed if and only if $\partial A \subseteq A$;
4. A is open if and only if $\partial A \subseteq A^c$;
5. $\partial(A \cap B) \subseteq \partial A \cup \partial B$. The equality holds if $\overline{A} \cap \overline{B} = \emptyset$;
6. $\partial(\operatorname{int} A) \subseteq \partial A$.

Thank You!

We'd love your questions and feedback.

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(Lectures, walkthroughs, and course updates)

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References

[1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.